

## Matrix Problem Bank

### Legend

★ = Core Placement Problem (Mandatory)

Language-Agnostic DSA Problem Bank | Focus: 2D array manipulation and grid-based algorithmic thinking.

### Section 1: Matrix Internals and Basic Operations

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- ★ Create and Display an  $M \times N$  Matrix
- ★ Access an Element at Row  $i$ , Column  $j$
- ★ Row-Wise Traversal of a Matrix
- ★ Column-Wise Traversal of a Matrix
- ★ Update an Element at a Given Position

Insert a Row at a Given Position

Insert a Column at a Given Position

Delete a Row from a Matrix

Delete a Column from a Matrix

### Section 2: Matrix Traversal Patterns

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- ★ Primary Diagonal Traversal
- ★ Secondary Diagonal Traversal
- ★ Boundary (Perimeter) Traversal of a Matrix
- ★ Spiral Order Traversal

Zigzag (Snake Pattern) Traversal

Anti-Diagonal Traversal

### Section 3: Matrix Transformation

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- ★ Transpose of a Matrix
- ★ Rotate Matrix by 90 Degrees Clockwise
- Rotate Matrix by 90 Degrees Anti-Clockwise
- Rotate Matrix by 180 Degrees
- ★ Horizontal Flip (Reverse Each Row)
- Vertical Flip (Reverse Each Column)

### Section 4: Matrix Search Problems

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- ★ Search an Element in a Row-Wise and Column-Wise Sorted Matrix
- ★ Find Row with Maximum Number of 1s (Binary Matrix)
- Count the Number of 1s in a Sorted Binary Matrix
- ★ Search in a Row-Sorted Matrix (Binary Search per Row)
- Search in a Fully Sorted Matrix (Binary Search on Matrix)

### Section 5: Matrix Computation Problems

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- ★ **Sum of Each Row and Each Column**
- ★ **Sum of Primary and Secondary Diagonals**
- ★ **Matrix Addition and Subtraction**
- ★ **Matrix Multiplication**

Check if a Matrix is Symmetric

Check if a Matrix is an Identity Matrix

Scalar Multiplication of a Matrix

## Section 6: Special Matrix Problems

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- ★ **Set Matrix Zeroes (If Element is 0, Set Row and Column to 0)**

- ★ **Pascal's Triangle using Matrix**

Toeplitz Matrix Check

- ★ **Print Matrix in Diagonal Pattern**

Find the Saddle Point of a Matrix

Rotate Image In-Place without Extra Space

## Section 7: Grid-Based Problems (Graph on Matrix)

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- ★ **Number of Islands (Connected Components in Binary Matrix)**

- ★ **Flood Fill Algorithm**

- ★ **Shortest Path in Binary Matrix (BFS)**

Surrounded Regions (Capture Regions of O)

- ★ **Count Paths from Top-Left to Bottom-Right**

Minimum Cost Path in a Grid

Word Search in a 2D Grid (DFS + Backtracking)

## Section 8: Real-World Applications of Matrix

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- ★ **Digital Image Rotation — Rotate Pixel Matrix by 90 Degrees**

- ★ **Image Blur Filter — Replace Each Pixel with Average of its Neighbours**

Conway's Game of Life — Simulate Next Generation of Cells

- ★ **Tic-Tac-Toe Board — Check Winner after Each Move**

Chess Board Validation — Check Valid Positions for Knight Moves

- ★ **Seating Arrangement Checker — Find Empty Seats with No Adjacent Occupants**

Spreadsheet Cell Reference Resolution (Dependency Matrix)

Minesweeper Board Generator — Place Mines and Compute Adjacent Counts